



Faculty Guide:
Dr. Tushar Mote Sir

Says

- Energy consumption is difficult to track in real time.
- Monthly electricity bills do not show where energy is being wasted.
- Equipment abnormalities can be difficult to identify.

Thinks

- Can abnormal energy usage be detected early?
- Can future electricity demand be predicted?
- Can AI help reduce energy wastage?

User

Feels

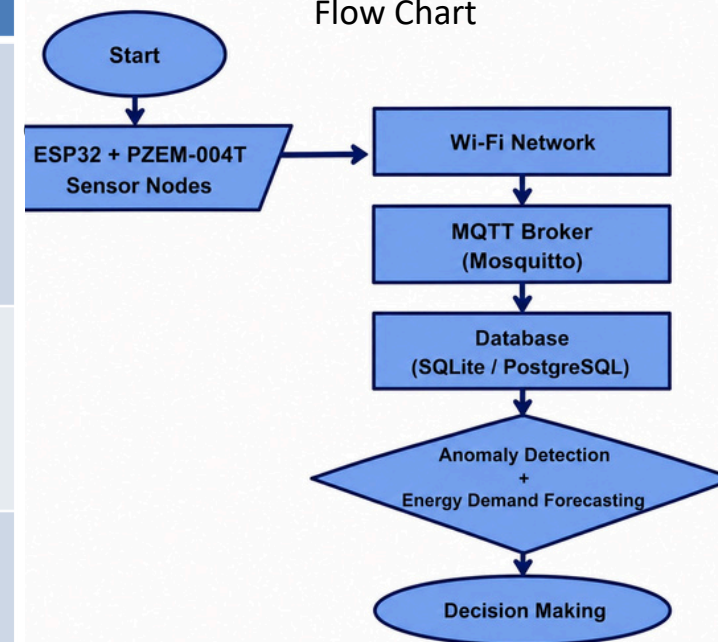
- Concerned about increasing electricity costs.
- Needs better visibility and control over energy consumption.
- Wants reliable information for energy management.

Does

- Monitors energy consumption across campus facilities.
- Checks energy usage patterns.
- Uses dashboard insights for energy management decisions.

Requirement	Analysis	Proposed Solution
Real-Time Monitoring	Traditional manual readings and monthly electricity bills do not provide real-time energy visibility.	ESP32 with PZEM-004T / ACS712 sensors continuously monitor voltage, current, power, and energy consumption.
Wireless Communication	Energy data from multiple campus locations needs reliable wireless transmission.	MQTT over Wi-Fi transmits sensor readings to a centralized backend server.
AI-Based Analysis	Abnormal energy consumption and equipment faults need to be detected early.	Machine learning models perform anomaly detection, demand forecasting, and consumption profiling.

Flow Chart



Problem statement

Campus facilities such as classrooms, laboratories, libraries, and server rooms consume significant amounts of electricity. Traditional monthly bills and manual meter readings provide no real-time visibility into energy consumption, wasteful usage, or equipment anomalies. The project aims to monitor electricity consumption across multiple campus locations, detect abnormal energy usage, and predict future electricity demand to reduce avoidable costs and energy wastage.

Proposed Solution

The proposed system uses ESP32-based wireless sensor nodes with current and voltage sensors to continuously monitor electricity consumption. Sensor data is transmitted through Wi-Fi and MQTT to a FastAPI backend and centralized database. AI/ML models analyze the data for anomaly detection, energy demand forecasting, and building-wise consumption profiling. A Streamlit dashboard provides real-time analytics, alerts, and actionable insights.

Scope and Feasibility

The system focuses on real-time energy monitoring across classrooms, laboratories, libraries, and server rooms. It is technically feasible using ESP32 sensors, MQTT, FastAPI, databases, machine learning, and Streamlit. The estimated implementation cost is ₹9,000–₹11,000. Tcommunication, cloud deployment, mobile applications, TinyML, and solar energy monitoring.